

Errors and Precision Lab

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Use float and appropriate functions from math library to evaluate square root etc in all the examples below

1. Consider two numbers 0.1 and 0.2. Add them and check if the sum is equal to 0.3
2. Sum 0.1 10^5 times and print out the sum and compare with the expected value
3. Check if the associativity $(a + b) + c = a + (b + c)$ is valid for the following numbers $a = 10^{20}$, $b = -10^{20}$, $c = 1$
4. Calculate the roots of the quadratic equation $x^2 + 10^4x + 1 = 0$
5. Calculate the function $f(x)$ for different values of x

$$f(x) = \sqrt{x+1} - \sqrt{x}$$

Reason your answers and see if you can improve the calculations

6. Calculate and plot the error in computing derivative of $\sin(x)$ for different values of Δx at $x = 1$. The definition of the numerical derivative is given by

$$\frac{d}{dx} \sin(x) = \frac{\sin(x + \Delta x) - \sin(x)}{\Delta x}$$

Compare it with the true value of the derivative ($\frac{d\sin(x)}{dx} = \cos(x)$ at $x = 1$)

7. The Harmonic series ($S = \sum_{i=1}^{i=\infty} \frac{1}{i}$) diverges. Can you check if it really diverges in C code